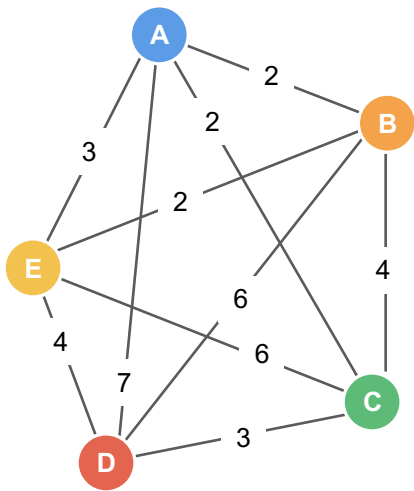


旅行商问题示例



任务：寻找从节点 A 出发的最短哈密顿回路 (Hamiltonian Cycle)

PlanGraph流程示例

问题智能体

求从节点 A 出发并的最短环路

规划智能体

伪代码规划:

1. Build graph
2. Enumerate cycles
3. Compute cost
4. Select minimum

检索智能体

add_edges()

nx.Graph()

TSP()

编码智能体

生成Python代码

结果

A → B → E → D → C → A 花费 = 14

推理想智能体

分析与反馈

执行 & 验证

运行与检查



```
import networkx as nx
from networkx.algorithms import TSP

G = nx.Graph()
G.add_weighted_edges_from([
    ('A', 'B', 2), ('A', 'C', 2),
    ('A', 'D', 7), ('A', 'E', 3),
    ('B', 'C', 4), ('B', 'D', 6),
    ('B', 'E', 2), ('C', 'D', 3),
    ('C', 'E', 6), ('D', 'E', 4)
])

tour = traveling_salesman_problem(G, cycle=True)
print(tour)
```